SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

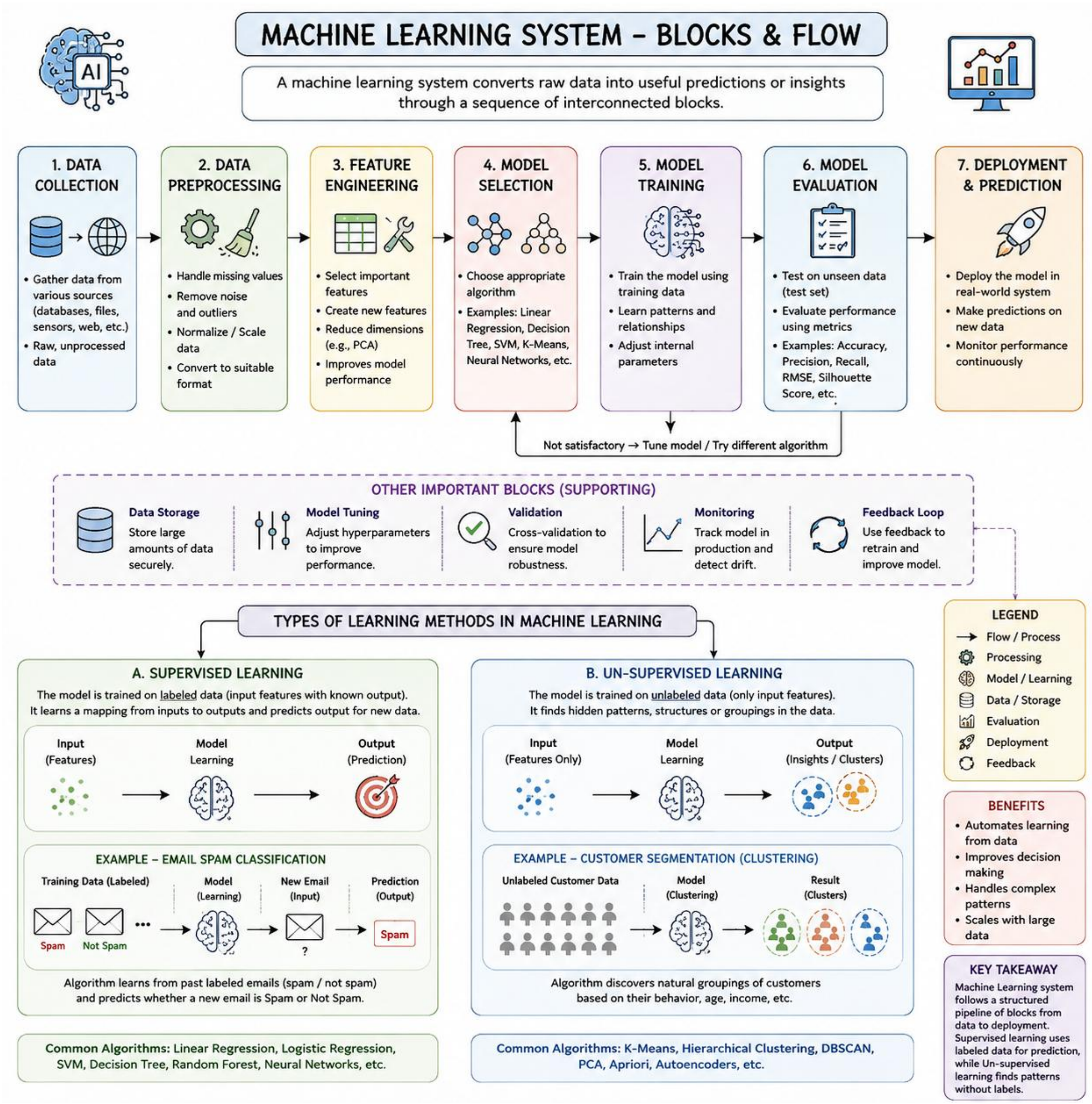
CONCEPT MAPPING

Code of Conduct:

I Cheredla Jasvina(192324026)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

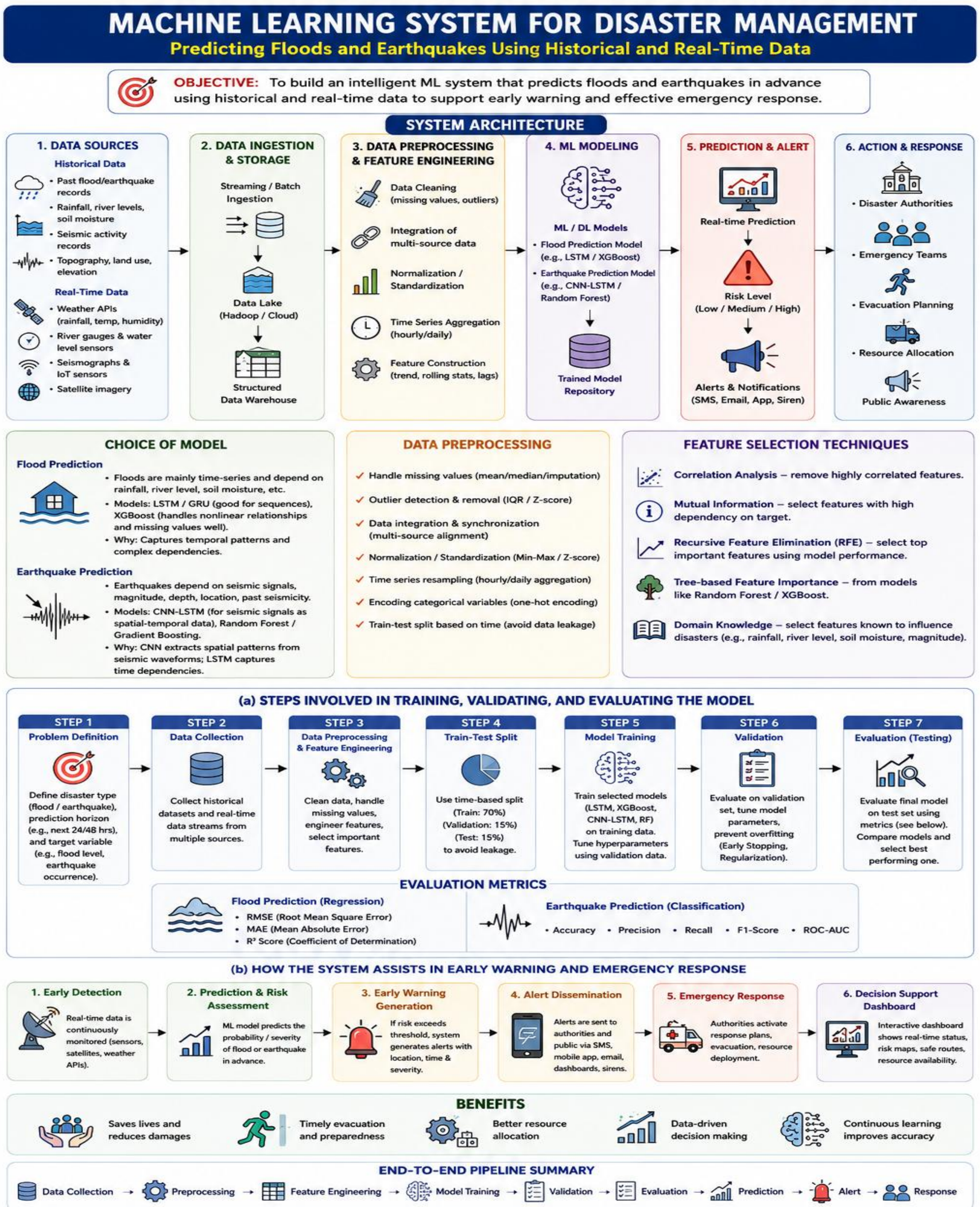
1.With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme.



2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques.

(a) Describe the steps involved in training, validating, and evaluating the model.

(b) Explain how the system can assist in early warning and emergency response.



3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis.

Dataset:

Example	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Find-S Algorithm – Step-by-Step Solution

Find-S Algorithm: Finds the most specific hypothesis that is consistent with all positive training examples.

Attributes (Features)
Outlook, Temperature, Humidity, Wind

Dataset: Play Tennis

Example No.	Outlook	Temperature	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rainy	Mild	High	Weak	Yes
5	Rainy	Cool	Normal	Weak	Yes
6	Rainy	Cool	Normal	Strong	No
7	Overcast	Mild	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No

Legend
Ø : Most specific value (no information)
? : Any value (generalized)

Important Points
- Start with the most specific hypothesis.
- Consider only positive examples.
- Generalize minimally when a positive example has a different value.

Initialization
The Find-S algorithm starts with the most specific hypothesis.

h_0 = (Ø, Ø, Ø, Ø)
(Outlook, Temperature, Humidity, Wind)

Step-by-Step Hypothesis Updates (Ignore Negative Examples)

Step	Example No.	Play Tennis	Training Example (Outlook, Temperature, Humidity, Wind)	Comparison & Update	Updated Hypothesis h
0 (Initial)	–	–	–	Start with most specific hypothesis	(Ø, Ø, Ø, Ø) (Most specific)
1	1	No	(Sunny, Hot, High, Weak)	Negative example → Ignore	(Ø, Ø, Ø, Ø)
2	2	No	(Sunny, Hot, High, Strong)	Negative example → Ignore	(Ø, Ø, Ø, Ø)
3	3	Yes	(Overcast, Hot, High, Weak)	First positive example → Set h to this example	(Overcast, Hot, High, Weak)
4	4	Yes	(Rainy, Mild, High, Weak)	Outlook: Overcast ≠ Rainy → ? Temperature: Hot ≠ Mild → ? Humidity: High = High → High Wind: Weak = Weak → Weak	(?, ?, High, Weak)
5	5	Yes	(Rainy, Cool, Normal, Weak)	Outlook: ? → ? Temperature: ? → ? Humidity: High ≠ Normal → ? Wind: Weak = Weak → Weak	(?, ?, ?, Weak)
6	6	No	(Rainy, Cool, Normal, Strong)	Negative example → Ignore	(?, ?, ?, Weak)
7	7	Yes	(Overcast, Mild, Normal, Strong)	Outlook: ? → ? Temperature: ? → ? Humidity: ? → ? Wind: Weak ≠ Strong → ?	(?, ?, ?, ?)
8	8	No	(Sunny, Mild, High, Weak)	Negative example → Ignore	(?, ?, ?, ?)

Final Maximally Specific Hypothesis

h = (?, ?, ?, ?)

This hypothesis is consistent with all positive examples (Examples 3, 4, 5, and 7).

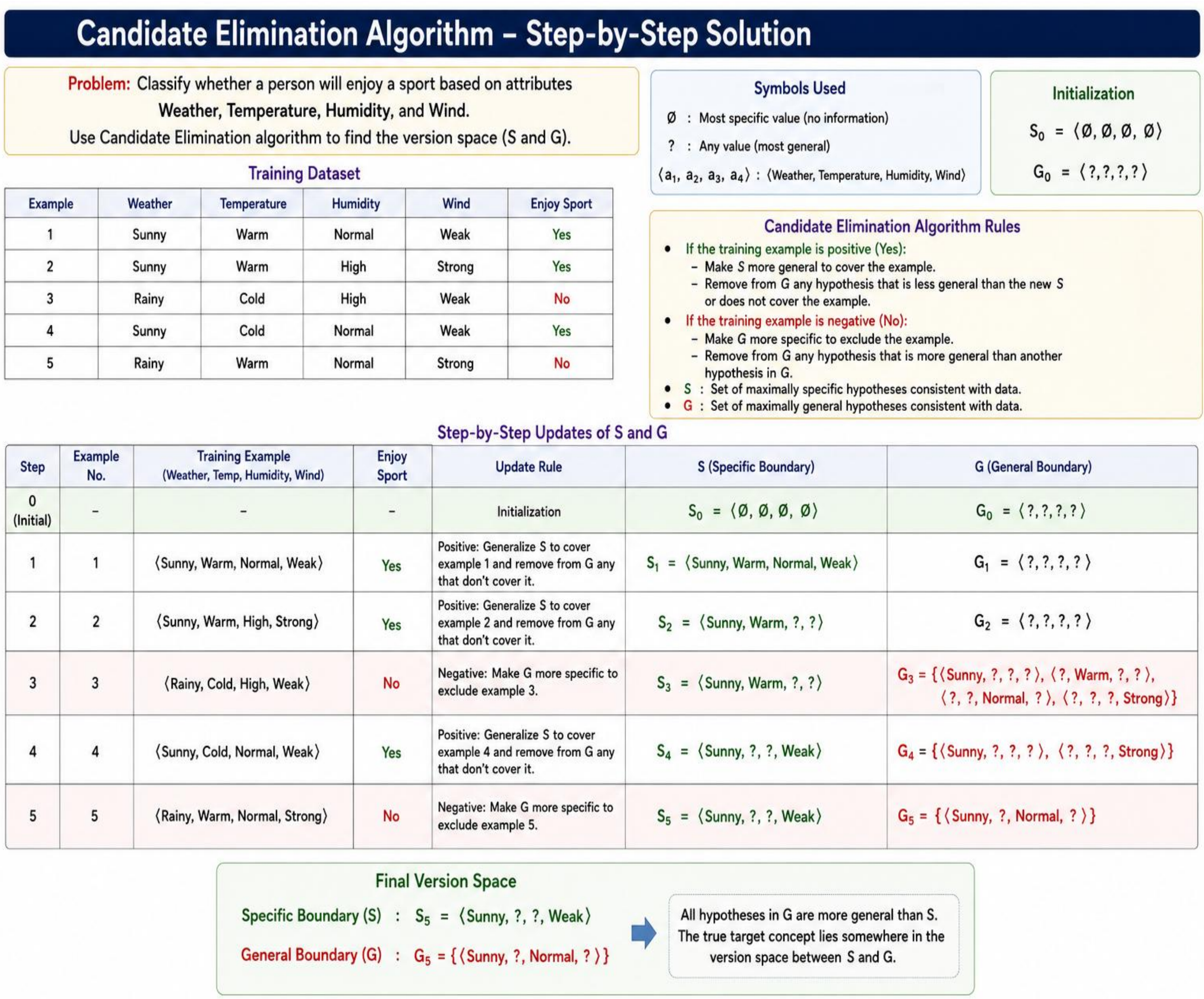
Conclusion
All attributes were generalized to "?" because they had different values in the positive examples.
Hence, the maximally specific hypothesis is (?, ?, ?, ?).

4. classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after

processing each training example.

Dataset:

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No



Candidate Elimination Algorithm – Step-by-Step Solution

Problem: Classify whether a person will enjoy a sport based on attributes **Weather, Temperature, Humidity, and Wind.**
Use Candidate Elimination algorithm to find the version space (S and G).

Symbols Used

- Ø : Most specific value (no information)
- ? : Any value (most general)
- $\langle a_1, a_2, a_3, a_4 \rangle$: ⟨Weather, Temperature, Humidity, Wind⟩

Initialization

$S_0 = \langle Ø, Ø, Ø, Ø \rangle$

$G_0 = \langle ?, ?, ?, ? \rangle$

Training Dataset

Example	Weather	Temperature	Humidity	Wind	Enjoy Sport
1	Sunny	Warm	Normal	Weak	Yes
2	Sunny	Warm	High	Strong	Yes
3	Rainy	Cold	High	Weak	No
4	Sunny	Cold	Normal	Weak	Yes
5	Rainy	Warm	Normal	Strong	No

Candidate Elimination Algorithm Rules

- If the training example is positive (Yes):
  - Make *S* more general to cover the example.
  - Remove from G any hypothesis that is less general than the new S or does not cover the example.
- If the training example is negative (No):
  - Make G more specific to exclude the example.
  - Remove from G any hypothesis that is more general than another hypothesis in G.
- S : Set of maximally specific hypotheses consistent with data.
- G : Set of maximally general hypotheses consistent with data.

Step-by-Step Updates of S and G

Step	Example No.	Training Example (Weather, Temp, Humidity, Wind)	Enjoy Sport	Update Rule	S (Specific Boundary)	G (General Boundary)
0 (Initial)	–	–	–	Initialization	$S_0 = \langle Ø, Ø, Ø, Ø \rangle$	$G_0 = \langle ?, ?, ?, ? \rangle$
1	1	⟨Sunny, Warm, Normal, Weak⟩	Yes	Positive: Generalize S to cover example 1 and remove from G any that don't cover it.	S_1 = ⟨Sunny, Warm, Normal, Weak⟩	$G_1 = \langle ?, ?, ?, ? \rangle$
2	2	⟨Sunny, Warm, High, Strong⟩	Yes	Positive: Generalize S to cover example 2 and remove from G any that don't cover it.	S_2 = ⟨Sunny, Warm, ?, ?⟩	$G_2 = \langle ?, ?, ?, ? \rangle$
3	3	⟨Rainy, Cold, High, Weak⟩	No	Negative: Make G more specific to exclude example 3.	S_3 = ⟨Sunny, Warm, ?, ?⟩	G_3 = {⟨Sunny, ?, ?, ?⟩, ⟨?, Warm, ?, ?⟩, ⟨?, ?, Normal, ?⟩, ⟨?, ?, ?, Strong⟩}
4	4	⟨Sunny, Cold, Normal, Weak⟩	Yes	Positive: Generalize S to cover example 4 and remove from G any that don't cover it.	S_4 = ⟨Sunny, ?, ?, Weak⟩	G_4 = {⟨Sunny, ?, ?, ?⟩, ⟨?, ?, ?, Strong⟩}
5	5	⟨Rainy, Warm, Normal, Strong⟩	No	Negative: Make G more specific to exclude example 5.	S_5 = ⟨Sunny, ?, ?, Weak⟩	G_5 = {⟨Sunny, ?, Normal, ?⟩}

Final Version Space

Specific Boundary (S) : S_5 = ⟨Sunny, ?, ?, Weak⟩

General Boundary (G) : G_5 = {⟨Sunny, ?, Normal, ?⟩}

All hypotheses in G are more general than S. The true target concept lies somewhere in the version space between S and G.